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Class 9 Biology Chapter 8

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Unit 8

Plant physiology

QUESTION NO 1: What is plant physiology?

ANSWER: PLANT PHYSIOLOGY: Plant Physiology is the branch of Botany which deal with the various questions related to the function of plant body such as how plants grow from seed to the whole body how plant absorb water and mineral and transport them to the leave.

TOPIC 8.1

QUESTION NO 1: What are nutrients?

Answer: NUTRIENTS: Nutrients are chemical substances in food needed for energy, growth, repair, and proper body functioning.

QUESTION NO 2. What is nutrition?

Answer: NUTRITION: Nutrition is the process by which organisms obtain and use food for energy, growth, and maintenance of life.

QUESTION NO 3. What is nutrition in plants?

Answer: Nutrition in plants is the process of making food using carbon dioxide, water, and sunlight through photosynthesis. It is called autotrophic nutrition.

TOPIC 8.1.1

QUESTION NO 1: What are the type of plant nutrients?

ANSWER: There are 2 type of plant nutrient

- Macronutrients
- Micronutrients

QUESTION NO 2: Difference between Macronutrients and Micronutrients?

ANSWER:

Feature	Macronutrients	Micronutrients
Required Amount	Needed in large amounts	Needed in small amounts

Feature	Macronutrients	Micronutrients
Function	Help in major growth and development processes	Help in specific physiological and enzyme functions
Examples	Nitrogen (N), Phosphorus (P), Potassium (K), etc.	Iron (Fe), Zinc (Zn), Copper (Cu), etc.

QUESTION NO 3: State that Nitrogen is important in protein synthesis?

ANSWER:

IMPORTANCE OF NITROGEN IN PROTEIN SYNTHESIS:

Nitrogen is a Macronutrient in plant Nitrogen is essential for protein synthesis because it is a key component of amino acids, which are the building blocks of proteins. It is also found in nucleic acids (DNA and RNA) that carry the genetic instructions for making proteins.

- **Sources of Nitrogen**

Dead organic matter

Nitrogen fertilisers

- **Uses of Nitrogen**

Help plant make amino acid

Needed for making nucleotides

Important for producing chlorophyll

Help in protein synthesis

- **Nitrogen deficiency**

Growth retardation (slows plant production)

Cause chlorosis i.e., yellowing of plant leaves

QUESTION NO 4: State the IMPORTANCE OF MAGNESIUM IN CHLOROPHYLL FORMATION?

ANSWER:

IMPORTANCE OF MAGNESIUM IN CHLOROPHYLL FORMATION:

Magnesium is Macronutrient in plant and it is Magnesium is essential for chlorophyll formation as it is the central atom in the chlorophyll molecule

- **Sources of Magnesium**

Dead plant derbies

Dolomitic limestone or Magnesium fertilisers

Uses of Magnesium

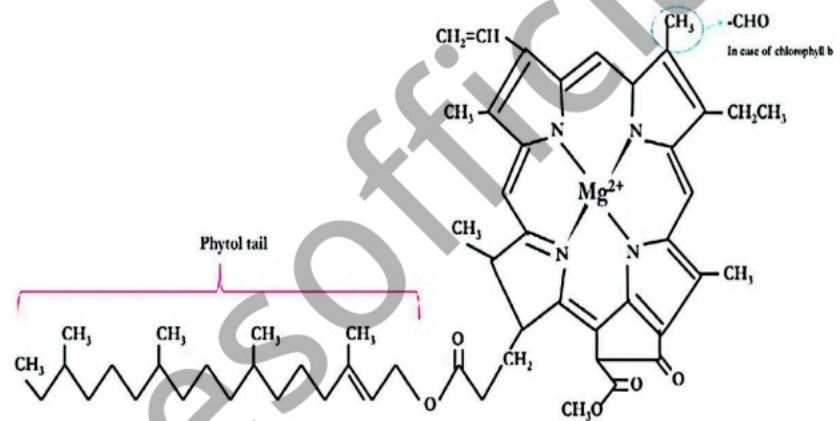
It helps in absorbing sunlight during photosynthesis, allowing plants to convert light energy into chemical energy.

- **Magnesium deficiency**

Without enough magnesium, plants cannot produce sufficient chlorophyll, leading to yellowing of leaves (chlorosis) and reduced growth.

QUESTION NO 5: Sketch the molecular structure of chlorophyll?

ANSWER: Molecular structure of chlorophyll:



TOPIC 8.2

QUESTION NO 1: Why do plant need to transport?

ANSWER:

- to. Move water and nutrients from soil to different parts of plants
- to carry food made in leaves to other parts of plant
- to remove waste materials

Topic 8.2.1

QUESTION NO 1: What are the level of transport in plant?

There are 3 levels of transport in plant

1st level: move material from outside to inside the plant body

2nd level: the materials are transport within the body

3rd level: the material move from inside to outside

Topic 8.2.2

QUESTION NO 1: What is the Structure of root and root hair?

ANSWER:

Structure of root and root hair

ROOT: The root is the part of a plant that grows underground. It holds the plant in the soil, absorbs water and minerals, and sometimes stores food.

ROOT HAIR: Root hairs are very small, thin extension of the root. They help the root to absorb more water and minerals from the soil by increasing the surface area.

Root in dicot plant

They have tap root which consists of primary root that bears several secondary Roots

Roots in monocot plant

They have fibrous adventitious Roots that show no difference of primary, secondary and tertiary roots



QUESTION NO 2: What are the internal structures of root and root hair?

ANSWER:

Internal structures of root and root hair

1. Epiblema (Epidermis):

Outermost layer with root hairs that absorb water and minerals.

2. Cortex:

Loosely packed cells that store food and allow water movement inward.

3. Endodermis:

Inner boundary of cortex that controls entry of water and minerals.

4. Pericycle:

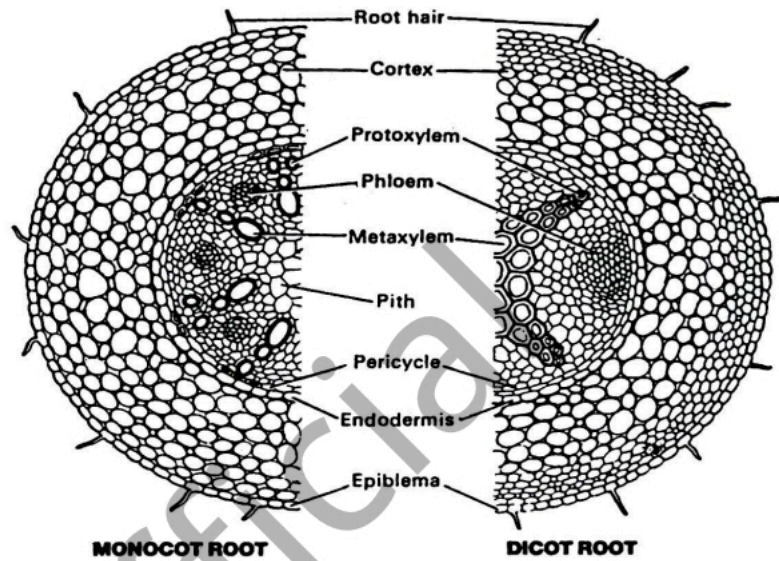
Layer inside endodermis that forms lateral roots.

5. Vascular Bundle (Xylem & Phloem):

Xylem carries water/minerals; phloem carries food.

6. Pith:

Central part that stores food; large in monocots, small or absent in dicots.



QUESTION NO 3: Difference b/w monocot and dicot root?

ANSWER:

Feature	Monocot Root	Dicot Root
Type of Root	Fibrous root system	Taproot system
Main Root	No main root; all roots are same size	One big main root with small side roots
Shape	Many small roots grow together	One thick root with tiny branches
Soil Depth	Grows in the top part of soil	Grows deep down into the soil
Examples	Wheat, Rice, Maize	Pea, Bean, Mustard

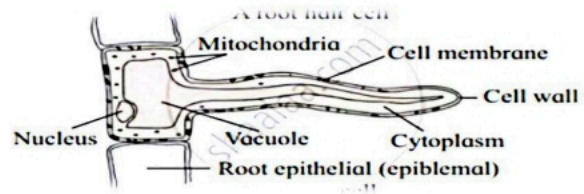
QUESTION NO 4: Difference b/w fibrous and taproot system?

ANSWER:

Feature	Fibrous Root System	Taproot System
Main Root	No main root; all roots are same size	One main root grows thick and deep
Appearance	Looks like a bunch of thin roots	One thick root with small side roots
Growth	Spreads near the surface of the soil	Grows deep into the soil
Plant Type	Found in monocot plants (e.g., wheat, rice)	Found in dicot plants (e.g., pea, mustard)

QUESTION NO 1: Draw the structure of root hair?

ANSWER: structure of root hair



QUESTION NO 2: How do minerals enter root cells?

Answer: Minerals enter through bulk flow with water, or by diffusion, facilitated diffusion, and active transport.

QUESTION NO 3: What structures on the roots absorb water and minerals, and how are they adapted for this?

ANSWER: Root hairs are the main structures that absorb water and minerals. They cover the branching roots and greatly increase surface area, allowing better contact with soil and more efficient nutrient and water uptake.

QUESTION NO 4: What are the two primary functions of plant roots, and how do they support the plant's survival?

ANSWER: The two primary functions of plant roots are:

1 Anchoring the plant: Roots hold the plant firmly in the soil, keeping it stable and upright so it doesn't fall over.

2 Absorbing water and nutrients: Roots take in water and essential minerals from the soil, which help the plant grow, make food, and stay healthy.

Together, these functions support the plant's survival by providing stability and supplying the materials needed for growth and energy.

QUESTION NO 5: How do plants absorb minerals from the soil, and what role does energy (ATP) play?

ANSWER: Plants absorb minerals through their root cells, but only when minerals are dissolved in soil water. The uptake happens in two ways:

➤ **Passive transport** – No energy is used; minerals move from high → low concentration naturally.

➤ **Active transport** – Requires energy (ATP); minerals move from low → high concentration, against the natural flow.

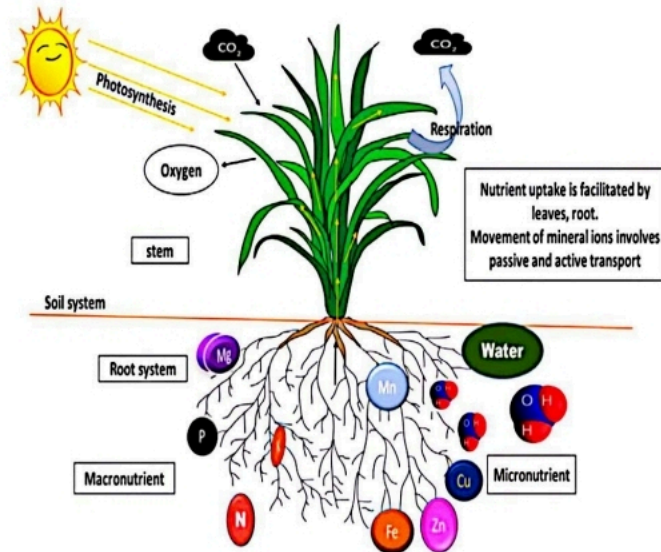
This method uses energy to help plants take in important minerals from the soil, even when there are only tiny amounts of those minerals.

QUESTION NO 6: What is facilitated diffusion in plants, and how does it work?

ANSWER: Facilitated diffusion in plants is the passive movement of minerals from high → low concentration through root cell membranes with the help of carrier proteins, without using energy (ATP)

QUESTION NO 7: How does active and passive transport of minerals occur in plants?

Answer: Mineral Uptake in Plants: Plants absorb minerals in dissolved form through their roots by two methods:



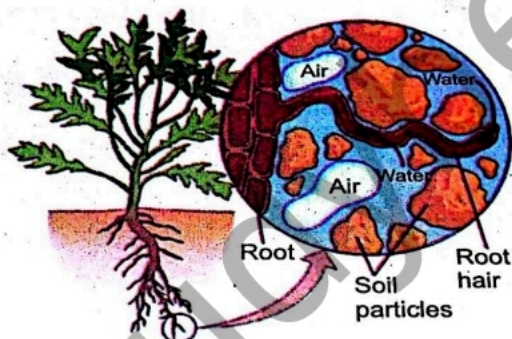
➤ **Passive Uptake** – Minerals move from high → low concentration by diffusion through plasmodesmata into the root cortex, endodermis, and pericycle, then enter the xylem and are pulled up by transpiration. In facilitated diffusion, carrier proteins in root cell membranes help minerals cross without energy use.

➤ **Active Uptake** – Uses ATP to move minerals from low → high concentration, against the gradient. This process is selective, depends on respiration, and lets plants absorb nutrients even when

scarce. Some ions may still enter by passive means when possible.

QUESTION NO 8: How does water absorb by root in plants?

ANSWER: Roots absorb water by a process called osmosis, which is a passive transport method. In passive transport, water moves without using energy, from a place where water is more concentrated to a place where it is less concentrated, through a partially permeable membrane.



water and minerals pass through easily.

⊖ the cell membrane is selectively permeable, which means it only lets certain substances enter the cell.

✓ This helps the plant control what goes inside the cell.

Types of Osmosis in Cells

◆ Endosmosis

✦ Water enters the cell through osmosis.

◆ Exosmosis

● Water leaves the cell through osmosis.

✦ the outer cells of roots have a cell wall that lets

QUESTION NO 1: Define transpiration and its types, importance and disadvantages of transpiration?

ANSWER:

TRANSPIRATION: Transpiration is the process of loss of water as water vapour from the aerial parts of a plant and this loss of water is called Transpiration

✦✦ Types of Transpiration ✦✦

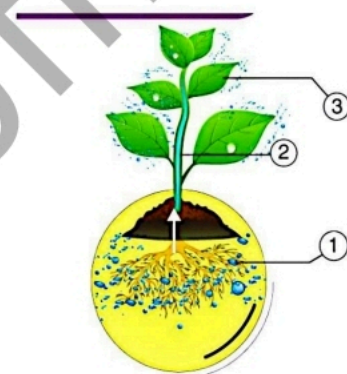
☞ **Stomatal Transpiration:** 90% of this loss occurs through the stomata present mainly in the leaves and some other aerial parts of the plant. This is called stomatal transpiration.

☞ **Cuticular Transpiration:** Some other amount of this absorbed water (7-9%) is lost from general body surface through the cuticle, called as cuticular transpiration.

☞ **Lenticular Transpiration:** A very little amount of water (less than 3%) is also lost through lenticels which are scar like regions present in the stem. This is called lenticular transpiration.

✦✦ Importance of Transpiration ✦✦

Transpiration the very important process for the plant as it is not only provides cooling to the plant but also help to absorb the water and dissolved minerals and to move them in upward direction to the leaves.



⚠️ Disadvantages of Transpiration ⚠️

Without transpiration plants are unable to survive. On the other hand, transpiration is generally regarded as disadvantageous as it shows great loss of absorbed water so, an unchecked excessive transpiration can lead to the wilting (loss of turgor) and ultimately death of the plant

QUESTION NO 2: Relation of Transpiration with Leaf Surface Area and Stomatal Opening/Closing?

ANSWER: ✦✦ TRANSPIRATION ✦✦: Transpiration mainly occurs through stomata, which are tiny pores present on the surface of leaves.

✦✦ Effect of Leaf Surface Area ✦✦

Plants with a large leaf surface area have more stomata.

More stomata allow more water vapour to escape, so these plants have a higher rate of transpiration.

Plants with narrow or reduced leaves have fewer stomata, so they have a lower rate of transpiration.

✦ Effect of Stomatal Opening and Closing ✦

When stomata are open, transpiration rate is high. When stomata are closed, transpiration rate is low.

✦ Day vs. Night Transpiration ✦

during the day, stomata are widely open, so transpiration is greater. At night, stomata are nearly closed, so transpiration is less.

TOPIC 8.2.5

QUESTION NO 1: What are the factors that affect the rate of transpiration?

ANSWER: Here are the some factors that affect the rate of transpiration in plant:

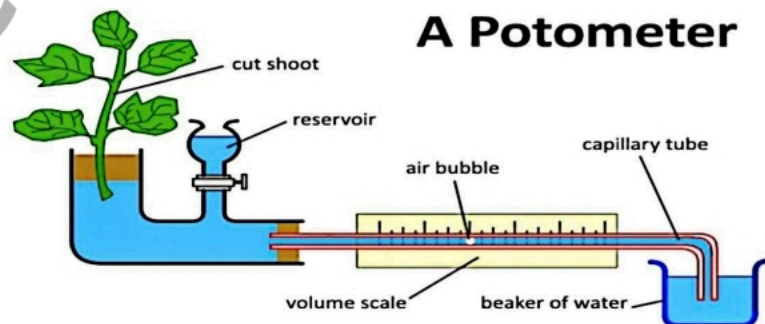
- 1 Temperature
- 2 wind
- 3 humidity

QUESTION NO 2: Name the device which is used to measure the rate of transpiration and its type?

ANSWER: A **potometer** is a device which is used to measure the rate of transpiration in plant by determining the amount of water which is absorbed by plant and also it is roughly equal to the water which is lost by transpiration

◆ TYPE ◆

- 1 bubble potometer
- 2 weight potometer



A Potometer

QUESTION NO 3: What are the other factors that affect the rate of transpiration?

ANSWER: Here are the some factors that affect the rate of transpiration:

- 1 Ability of soil water
- 2 Sunlight
- 3 Carbon dioxide concentration

QUESTION NO 3: What is the effect of temperature on the process of transpiration?

ANSWER: ◆ Effect of temperature on transpiration: ◆

ANSWER:

On hot sunny days, the air gets warmer and drier. This makes plants lose water faster through their leaves. If the temperature goes up by 10°C, they can lose about twice as much water. But when it gets extremely hot (around 40°C–45°C), plants close the tiny holes in their leaves to save water.

If the hot weather continues and the soil doesn't have enough water, the plant can't replace the water it loses. This makes the plant wilt, and if it goes on for too long, the plant can die.

QUESTION NO 4: What is the effect of wind and humidity on the process of transpiration

ANSWER: ♦ Effect of wind on transpiration: ♦

Wind is moving air. It makes plants lose water faster because it blows water away from the leaves quickly. This increases evaporation and transpiration. When there is no wind and the air is still, water leaves the leaves more slowly, so transpiration is less.

♦ Effect of humidity on transpiration: ♦

Humidity means how much water vapor is in the air. In dry air (low humidity), water leaves the plant's leaves faster because it can escape easily through the stomata. This makes the plant lose more water and increases transpiration.

In humid air (high humidity), there is already a lot of water in the air, so water leaves the leaves more slowly. This lowers the rate of transpiration.

TOPIC 8.2.6

QUESTION NO 1: Which tissue helps in the ascent of sap, and which theory explains the force behind it?

ANSWER: The xylem tissue carries water and minerals upward from roots to leaves, a process called the ascent of sap. The most widely accepted explanation for the force involved is the TACT theory (Transpiration pull, Adhesion, Cohesion, and Tension).

QUESTION NO 2: What is the passage for the ascent of sap, and which two xylem cells are important?

ANSWER: Xylem is the passage for ascent of sap. Its two important cells are:

✦✦ **Tracheids:** long, dead, pointed cells.

✦✦ **Vessels:** long, dead, thick-walled tubes

Together these help in water and mineral transport and support.

QUESTION NO 3: Explain the four factors of the TACT theory that together help move water and minerals upward in plants.

ANSWER: The TACT acronym stands for:

Transpiration Pull

Water evaporates from leaf stomata into the air.

Creates a pulling force that draws water upward from roots through the xylem.

Driven by solar energy—evaporation is the key trigger.

☐ **Adhesion**

Attraction between water molecules and polar cellulose walls of xylem cells.

Helps water climb against gravity.

Keeps water in xylem even when transpiration slows down.

Cohesion

Hydrogen bonds between water molecules create strong attraction.

Maintains a continuous column of water from roots to leaves.

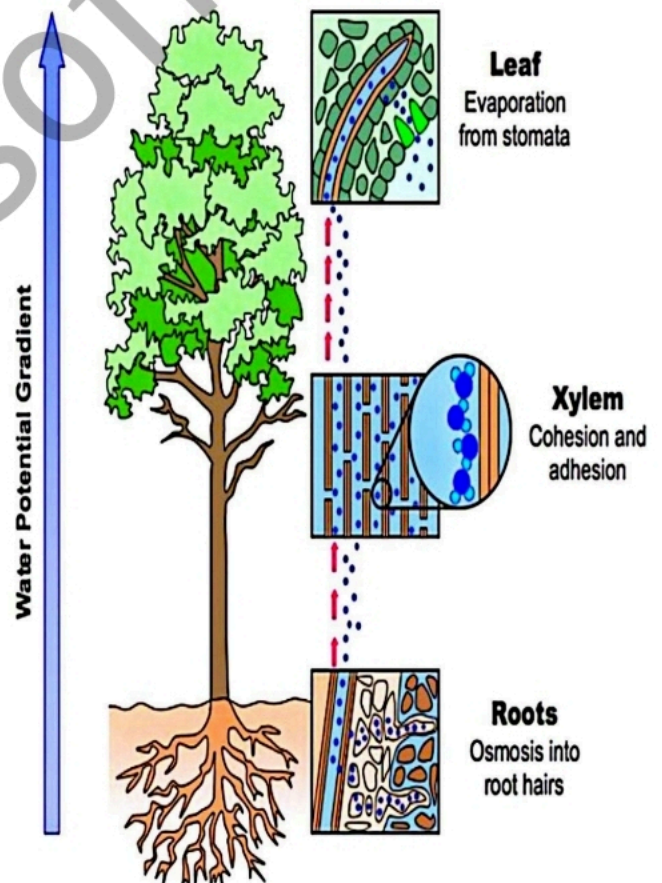
Tension

Caused by the pulling effect of transpiration at the top.

Keeps the water column tight and unbroken, like a stretched steel wire.

QUESTION NO 4: How TACT force work explain in short?

ANSWER: Imagine the water in the xylem like a steel wire—strong and unbroken. Hydrogen bonds between water molecules create **cohesion**, keeping them linked like a string. When water evaporates from the leaves during **transpiration**, it pulls the entire column of water upward. This upward movement, called **bulk flow**, is driven by **solar energy**, as the heat from the sun causes the water to evaporate and create the **transpiration pull**.



QUESTION NO 1: Explain the mechanism of food translocation by the theory of pressure flow mechanism? Explain in detail with diagram.

Answer: **TRANSCOLATION:** The transport of prepared food to different parts of the plant through the phloem tissues is known as translocation.

PRESSURE FLOW MECHANISM: The pressure flow mechanism is the process by which food (mainly sucrose) is transported through the phloem of plants from a region of high pressure called the source to a region of low pressure called the sink, due to differences in turgor pressure created by water movement.

PASSAGE OF TRASLOCATION: It is proved that translocation is occurs through phloem tissues which are complex .Sieve tube and companion cells are important components of phloem.

SIEVE TUBE: Sieve Tubes are long tube-like structures that transport organic solutes (like sucrose) throughout the plant. Since sieve tube elements lack nuclei.

COMPANION CELL: Companion cells next to them control their functions.

Together, sieve elements and companion cells act as conducting channels for moving organic solutes.

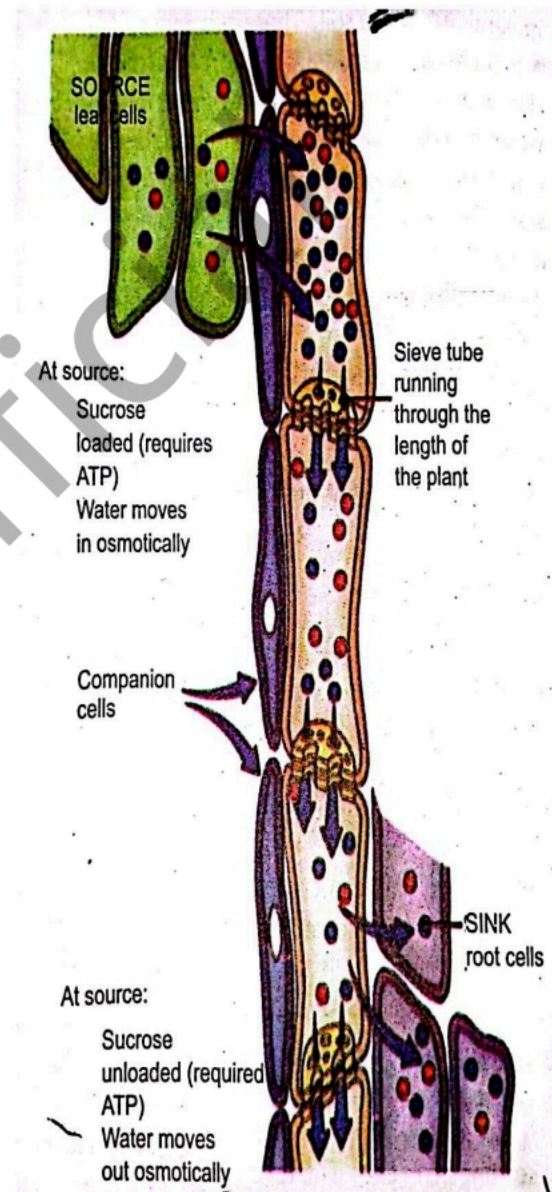
Direction of Translocation: This movement is not just up or down; it goes in various directions. We usually say that translocation goes from a "source" to a "sink."

SOURCE: A "source" supplies food (like leaves making food).

SINK: A "sink" uses food (like roots or storage organs).

Leaves are sources, fruits are sinks, and stems and roots can act as both.

Composition of Translocating Fluid (Phloem Sap): Phloem sap has about 10–25% solid material in it. Most of this solid material is sucrose (a type of sugar), along with a small amount of other organic substances.



🔑 **Mechanism of Translocation (Pressure Flow or Mass Flow Theory):** The most accepted idea for how organic solutes moves in plants is the "pressure flow" or "mass flow" theory. Here's how it works:

1. Sugars produced in source regions, like leaves or storage areas, are loaded into the phloem's sieve tube elements by companion cells.
2. This active transport increases the sugar level and lowers the water level in the phloem, making the phloem sap more concentrated with sugar.
3. Water moves into the phloem through osmosis from nearby xylem cells, increasing hydrostatic pressure (pressure from water) in the phloem cells. This pressure pushes the sugary solution away from the source (usually a leaf).
4. The pressure difference, from the high-pressure source to the lower-pressure sink, causes translocation.
5. When the phloem sap loses sugars and other substances, its water potential increases, so water moves back into the xylem tubes.

So by this process the food is transport in the plant body through phloem tissues.

Topic 8.3 + TOPIC 8.3.1 + TOPIC 8.3.2

QUESTION NO 1: Define gaseous exchange in plant give answer in detail?

ANSWER: GAS EXCHANGE IN PLANT: Gas exchange in plants is the process by which the whole plant swaps gases with the environment to meet its needs. It provides oxygen for cellular respiration, where complex food is broken down to release energy, and removes the carbon dioxide produced. Every cell does its own exchange as required. This mainly takes place through stomata in leaves and lenticels in stems, and it happens by diffusion.

🔑 **PATTERN OF GAS EXCHANGE IN LEAVES:**

Photosynthetic autotrophs like green plants change their gas exchange pattern between day and night because the processes of photosynthesis and respiration dominate at different times.

	IN DAY TIME	IN NIGHT TIME
INHALE	Carbon dioxide	Oxygen
EXHALE	Oxygen	Carbon dioxide

⚙️ **Daytime (Light Present):** During the day, plants do both photosynthesis (making food) and respiration. The speed of photosynthesis changes during the day because it depends mostly on light. Usually, photosynthesis happens faster than respiration, so plants need more carbon dioxide than what

respiration makes. They take the extra carbon dioxide from the air. Photosynthesis also makes more oxygen than respiration uses, so plants release the extra oxygen into the air.

🌙 **Night-time (No Light):** At night, plants act more like animals. They absorb oxygen and release carbon dioxide because there's no photosynthesis happening in the dark.

🌅 **Dawn/🌆 dusk (sunrise and sunset period):** During dawn and dusk, when there is little light, photosynthesis and respiration happen at the same rate. The carbon dioxide made in respiration is just enough for photosynthesis, and the oxygen from photosynthesis is used in respiration. At this time, there is no overall gas exchange with the environment. This is called the **compensation point of photosynthesis**.

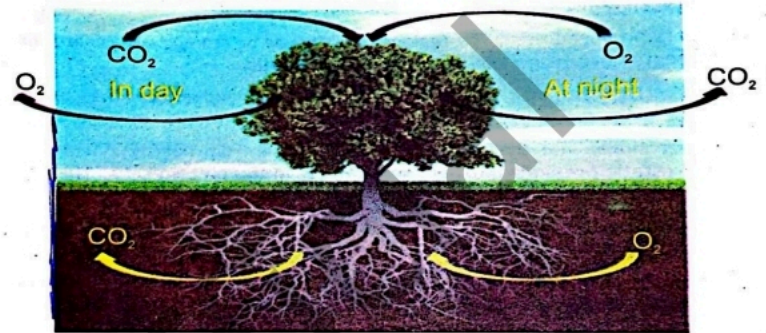


Figure: 8.10 Pattern of gas exchange in leaves and roots in day and night

🌱 **Pattern of Gas Exchange Pattern in Roots:**

Roots also exchange gases with soil through root epidermal cells which involves absorption and it remains same day and night. They take in oxygen and give out carbon dioxide both day and night because there's no photosynthesis happening in the roots

Topic 8.4

QUESTION NO 1: Define homeostasis in plant?

ANSWER: HOMEOSTASIS IN PLANTS: Homeostasis in plants refers to the adjustment of adaptation of plants to varying degrees of environmental temperatures and availability of water and salts in the soil.

Topic 8.4.1

Question no 1: Write the importance of homeostasis in plant?

ANSWER: importance of homeostasis in plant:

- 1 Homeostasis in plants is essential for their survival, growth, and reproduction
- 2 Homeostatic mechanisms help plants to stay balanced and alive in all types of weather.
- 3 Plants can use special defences to protect themselves from harm.

4 Homeostasis helps plants keep the right amount of water inside them.

5 Homeostasis helps plants deal with problems like less water, salty soil, diseases, and animals that eat them

6 it helps plants survive and stay healthy in changing environments

Topic 8.4.2

QUESTION NO 1: Describe the mechanism of adaptation of plant for the excretion?

ANSWER:

✦✦ Excretion in plants

unlike animals, excretion in plants is a wide process that includes removing any unwanted substance from the plant's body. Plants get rid of waste in a different way than animals. They produce different organic and inorganic compounds, which they may store for some time and remove when necessary. The following are some ways plants carry out excretion.

✦✦ Excretion of carbon dioxide and oxygen: Carbon dioxide is the waste of respiration which is excreted from plant body at night during gas exchange process. Similarly, oxygen is the waste of photosynthesis which is excreted in day during the same gas exchange process.

✦✦ Excretion of water: Water is produced as waste of both photosynthesis and respiration and it is also absorbed from soil. The excretion of water occurs through the plant body during the process of transpiration

✦✦ Excretion through roots: Some plants release chemicals from their roots into the soil to prevent neighbouring plants from competing for resources conifer trees are known from it

✦✦ Excretion through stem and branches: some trees keep waste chemicals in their stems and branches, especially in old parts that don't carry water anymore. For example, ebony trees store substances that make the middle of their wood very dark. Scientists see these substances as waste. The waste is not thrown out of the plant but kept in one place, like putting trash in a bin — it's still inside but gathered in one spot.

✦✦ Excretion through leaves: Plant cells have large vacuoles that can store both useful materials and waste. Sometimes, wastes collect in these vacuoles and form crystals. Leaves help in this process. When leaves store a lot of coloured compounds, they turn yellow. This yellow colour is not because of a loss of chlorophyll, like in chlorosis, but due to the build-up of these compounds. Such leaves usually fall from plants in the autumn season. In this way, leaves work as organs of excretion, so they are called **excretophores**.

QUESTION NO 1: Explain osmotic adjustment in plant? And its conditions and different groups.

ANSWER:

✦ **OSMOTIC ADJUSTMENT IN PLANT:** Osmotic adjustment, or **osmoregulation**, is how a plant keeps the right balance of water and dissolved substances in its body. This balance can lead to three conditions: hypotonic, hypertonic, and isotonic.

◆ Hypotonic condition – The surrounding solution has more water than the cell, so water moves into the cell, making it swell and firm also known turgid.

◆ Hypertonic condition – The surrounding solution has less water than the cell, so water moves out of the cell, making it shrink also known flaccid.

◆ Isotonic (ideal) condition – The water level inside and outside the cell is the same, so there's no net movement of water. This balance is found in plants natural environment.

✦ **OSMOTIC ADJUSTMENT IN DIFFERENT PLANTS GROUP:** they are categorized into 3 group based on water ability

◆ Hydrophytes: live in water rich environment.

◆ Mesophytes: live where water is moderately available.

◆ Xerophytes: live in extreme dry condition.

Table: 8.3 Comparison of osmotic adjustment in different groups of plant			
Features	Hydrophytes	Mesophytes	Xerophytes
Habitat	Aquatic	Terrestrial (moderate)	Terrestrial (severely dry)
Osmotic conditions of medium	Hypotonic	Nearly isotonic	Strong hypertonic
Problems faced by plant	Flooding	No drastic problem	Severe dehydration
Rate of transpiration	Very high rate	Medium rate	Very low rate